

Image and Video Generations

Lecture 4 Supplement: GLIDE / RealFill

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Recap: Classifier-Free Guidance (CFG)

How can we **enhance** the conditioning?

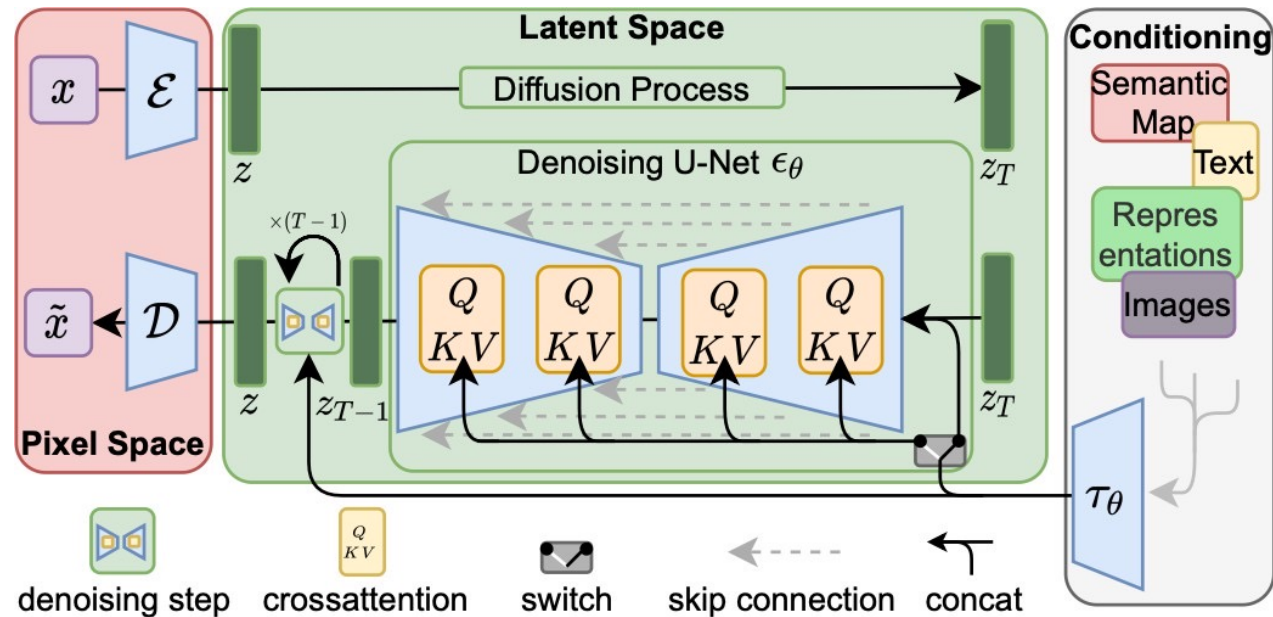
We can **extrapolate** the conditional noise $\hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t)$ from the null-condition noise $\hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)$:

$$\tilde{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) = (1 + \omega)\hat{\epsilon}_{\theta}(\mathbf{x}_t, \mathbf{y}, t) - \omega\hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)$$

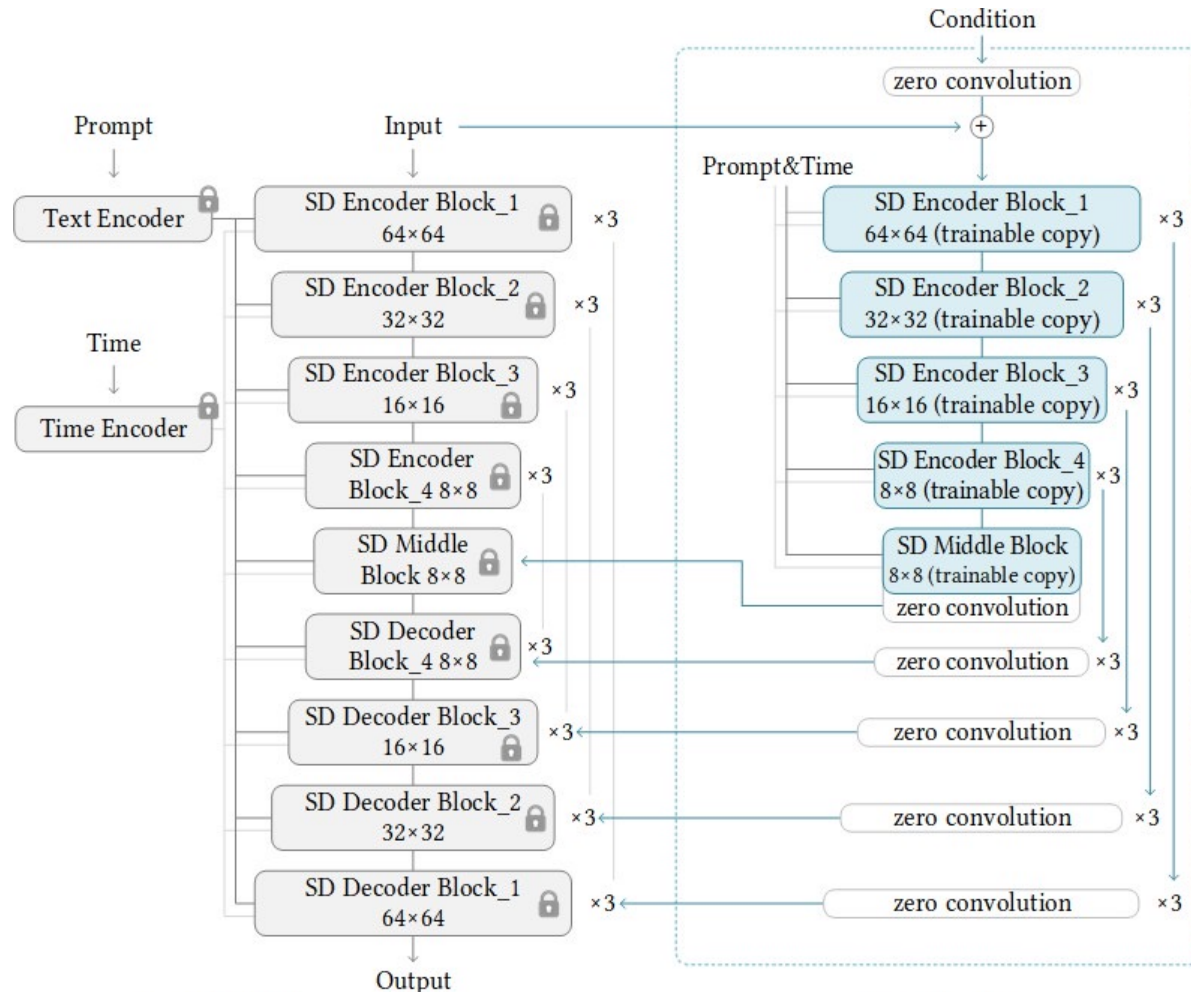
where $\omega \geq 0$.

Recap: Latent Diffusion

- The **noise predictor** is trained in the **latent space**.
- The input **condition** is incorporated into the **cross-attention** module.



Recap: ControlNet



(a) Stable Diffusion

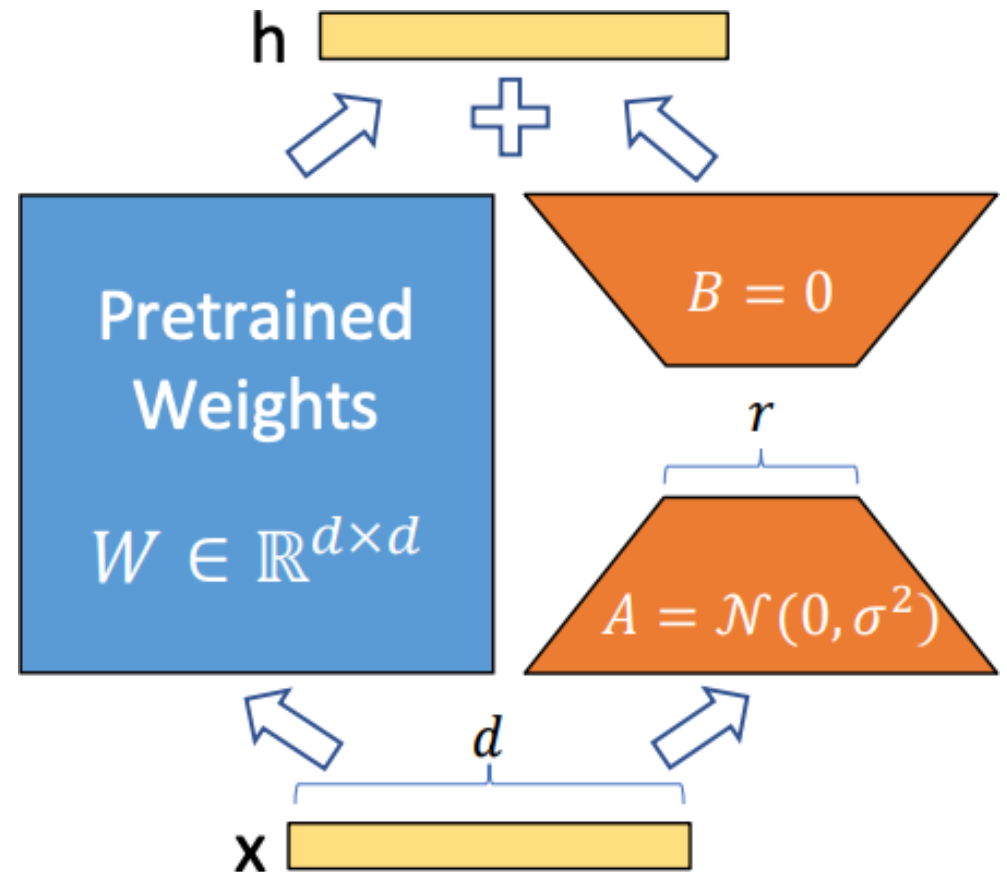
(b) ControlNet

1. Freeze the noise prediction network.
2. For the encoding of the conditional image, copy the pretrained encoder parameters while allowing them to be updated during fine-tuning.
3. Combine the encoded conditional image information with the noisy image using zero convolution.

Recap: Low-Rank Adaptation (LoRA)

Low-Rank Approximation:

Equivalent to having two MLP layers with a **low-dimensional intermediate output**.



Recap: Personalization

Input images



A [V] backpack in the Grand Canyon



A [V] backpack with the night sky



A [V] backpack in the city of Versailles



A wet [V] backpack in water

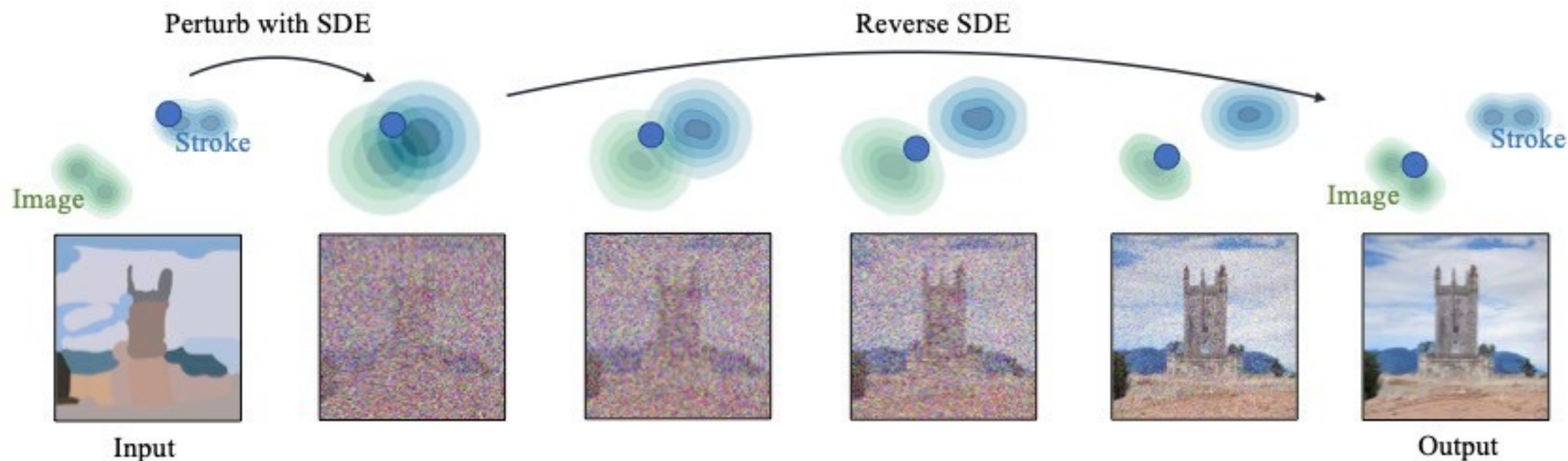


A [V] backpack in Boston

DreamBooth

Recap: SDEdit

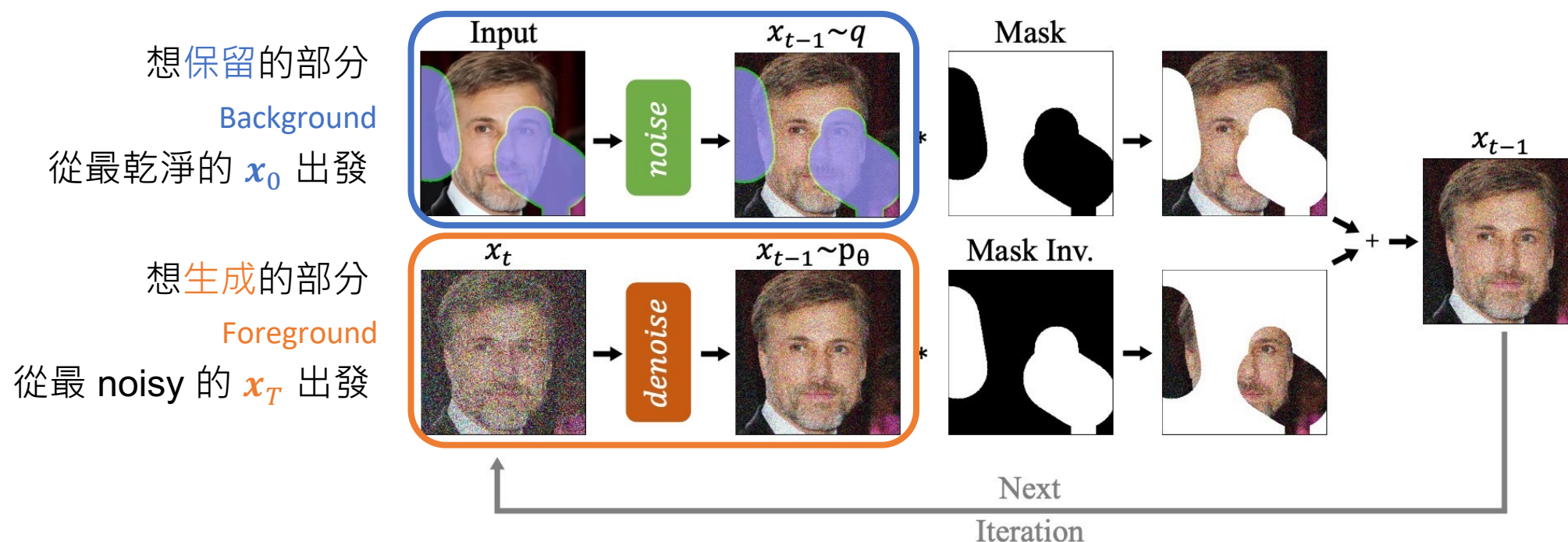
Perform the **forward** process for a bit and then **reverse** the process.



用 Noise 把 input 稍微淹沒，再讓 diffusion 去生成自然影像的細節

Recap: RePaint

Combine denoised **foreground** images (to be filled) and noisy **background** (to be fixed) images at each iteration of the reverse process.



Timeline

- [2021.05] Classifier Guidance
- [2021.12] Classifier-Free Guidance (CFG)
- [2021.12] **GLIDE** (ICML 2022)
 - **CFG + free-form text in pixel space** · finetune 一個特化的 inpainting model
- [2021.12] Latent Diffusion
 - CFG + free-form text in latent space
- [2022.08] Stable Diffusion/DreamBooth
- [2022.11] SD 2 Inpainting
- [2022.12] LoRA for SD/ControlNet
- [2023.09] **RealFill** (SIGGRAPH 2024)
 - Inpainting model + LoRA + reference images

GLIDE = Guided Language to Image Diffusion for Generation and Editing

Nichol et al., “GLIDE: Towards Photorealistic Image Generation and Editing with Text-Guided Diffusion Models”, ICML 2022.

Free-form Text Prompt -> Image Generation



"a hedgehog using a calculator"



"a corgi wearing a red bowtie and a purple party hat"



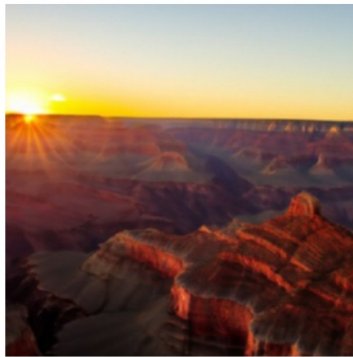
"robots meditating in a vipassana retreat"



"a fall landscape with a small cottage next to a lake"



"a surrealist dream-like oil painting by salvador dalí of a cat playing checkers"



"a professional photo of a sunset behind the grand canyon"

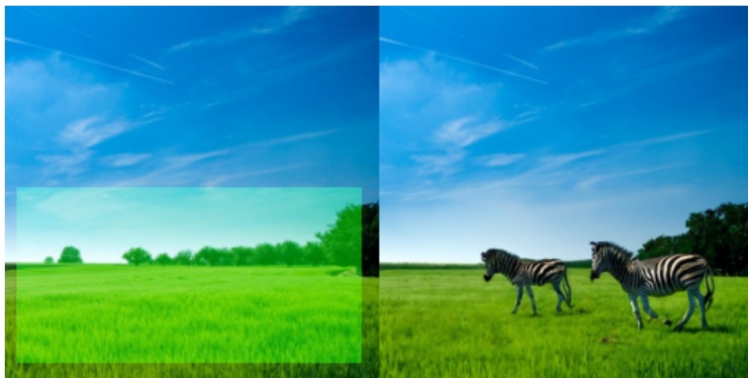


"a high-quality oil painting of a psychedelic hamster dragon"



"an illustration of albert einstein wearing a superhero costume"

Text-guided Inpainting



“zebras roaming in the field”



“a girl hugging a corgi on a pedestal”



“a man with red hair”



“a vase of flowers”

Related Work



Challenges

1. Classifier guidance 需要專門的 classifier，且 class set 固定
2. CLIP 沒看過 noise 或模糊影像 $\hat{x}_0$
3. Inpainting 只看得到加噪後的 context -> 邊界會不協調

“a corgi in a field”



Unnoised CLIP (+ aux losses)



Noised CLIP (+ upsampler)

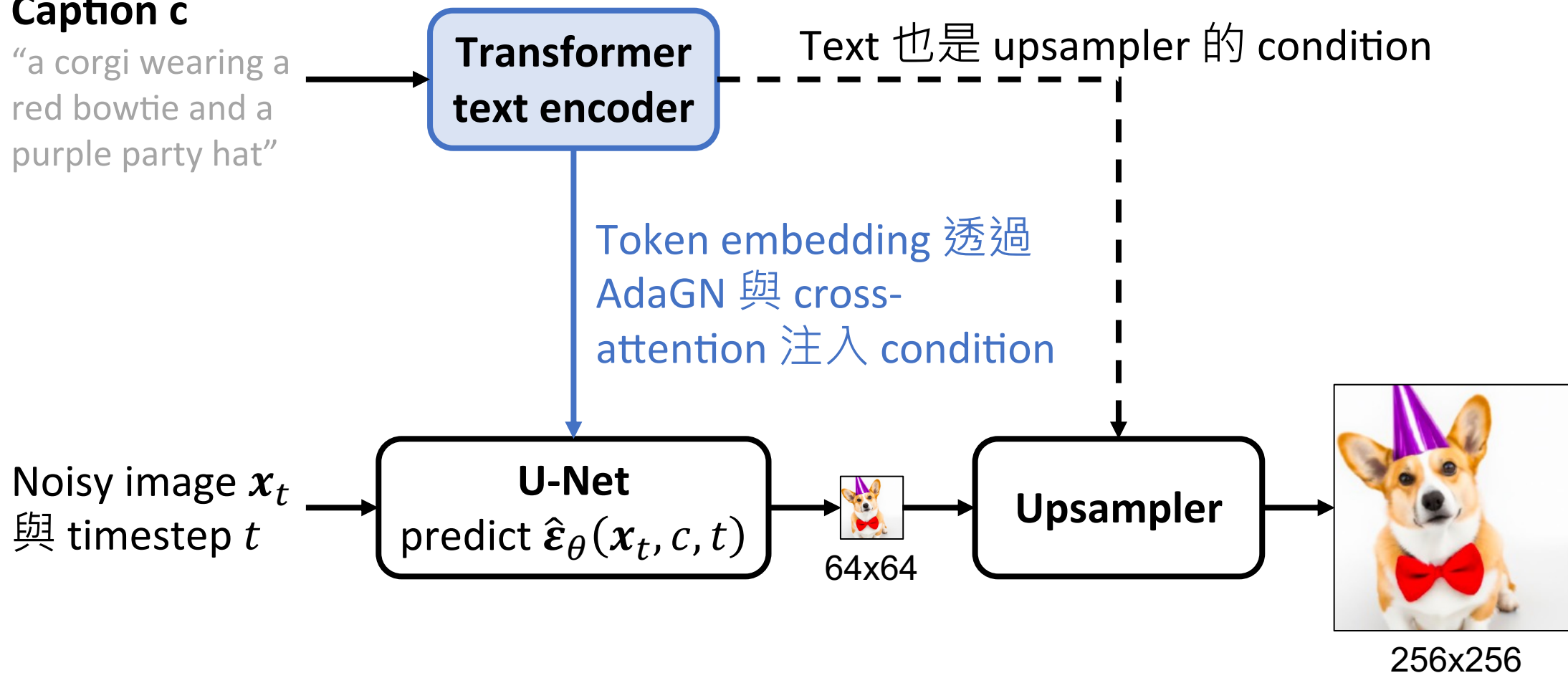


GLIDE

Method

Caption c

“a corgi wearing a red bowtie and a purple party hat”



CLIP guidance vs. CFG

CLIP guidance

$$\begin{aligned}\bar{\epsilon}_{\theta}(\mathbf{x}_t, t) \\ = \epsilon_{\theta}(\mathbf{x}_t, t) - \omega \sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{x}_t} (f(\mathbf{x}_t) \cdot g(c))\end{aligned}$$

需要額外訓練一個 noise-aware 的 CLIP

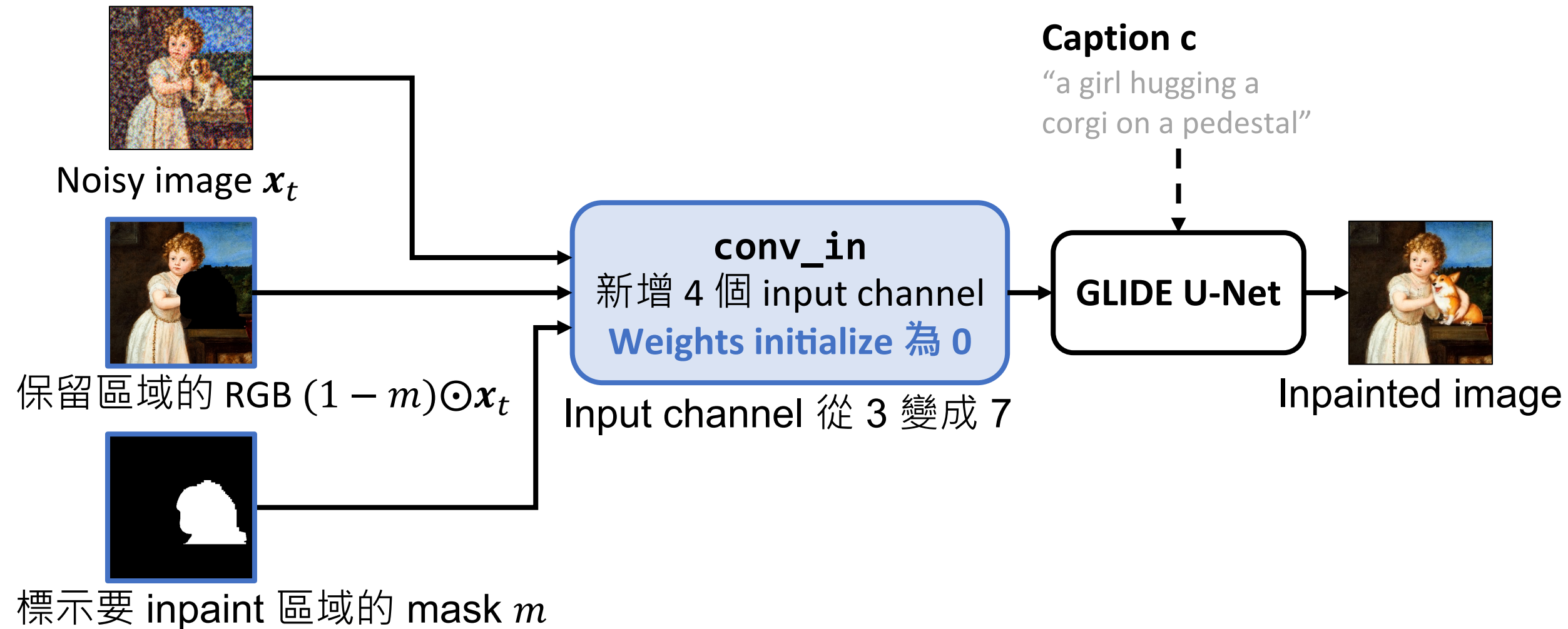
Classifier-free guidance

$$\begin{aligned}\tilde{\epsilon}_{\theta}(\mathbf{x}_t, c, t) \\ = \lambda \hat{\epsilon}_{\theta}(\mathbf{x}_t, c, t) - (1 - \lambda) \hat{\epsilon}_{\theta}(\mathbf{x}_t, \phi, t)\end{aligned}$$

不需額外 model，只使用 training data 的 knowledge

Method: Inpainting

多加入 input channel 來 finetune



Experiments: Human Evaluation

Table 1. Elo scores resulting from a human evaluation of unguided diffusion sampling, classifier-free guidance, and CLIP guidance on MS-COCO validation prompts at 256×256 resolution. For classifier-free guidance, we use scale 3.0, and for CLIP guidance scale 2.0. See Appendix A.1 for more details on how Elo scores are computed.

Guidance	Photorealism	Caption
Unguided	-88.6	-106.2
CLIP guidance	-73.2	29.3
Classifier-free guidance	82.7	110.9

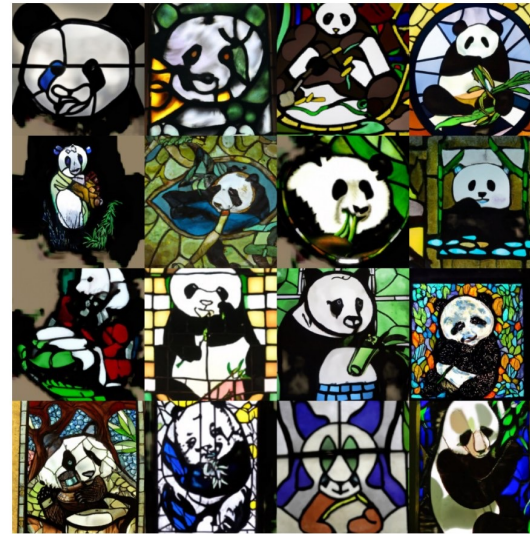
Experiments: Additional Samples



(a) DALL-E (Temp 0.85, CLIP reranked top 16 out of 512)



(b) GLIDE (Unguided)



(c) GLIDE (CLIP guidance, scale 2.0)



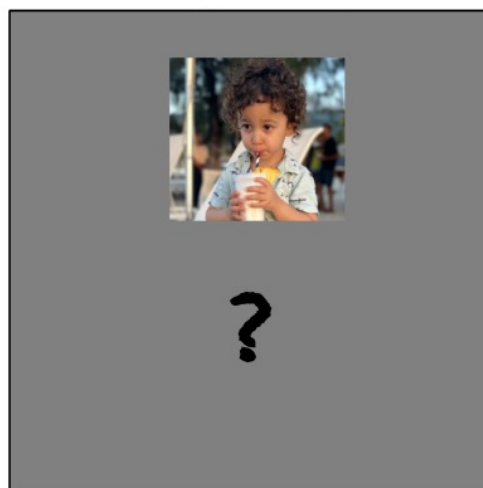
(d) GLIDE (Classifier-free guidance, scale 3.0)

Limitations

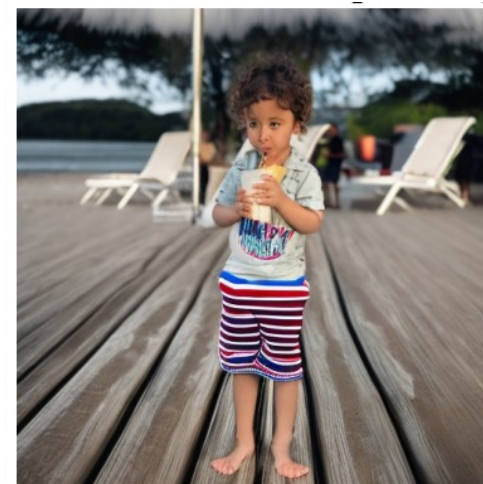
只能用文字描述要 inpaint 什麼 -> 有可能補出合理但不真實的內容



Reference images



Input image



Inpainted image

能不能用 reference images 讓模型記住這個場景，並 inpaint 出真正正確的內容？

RealFill

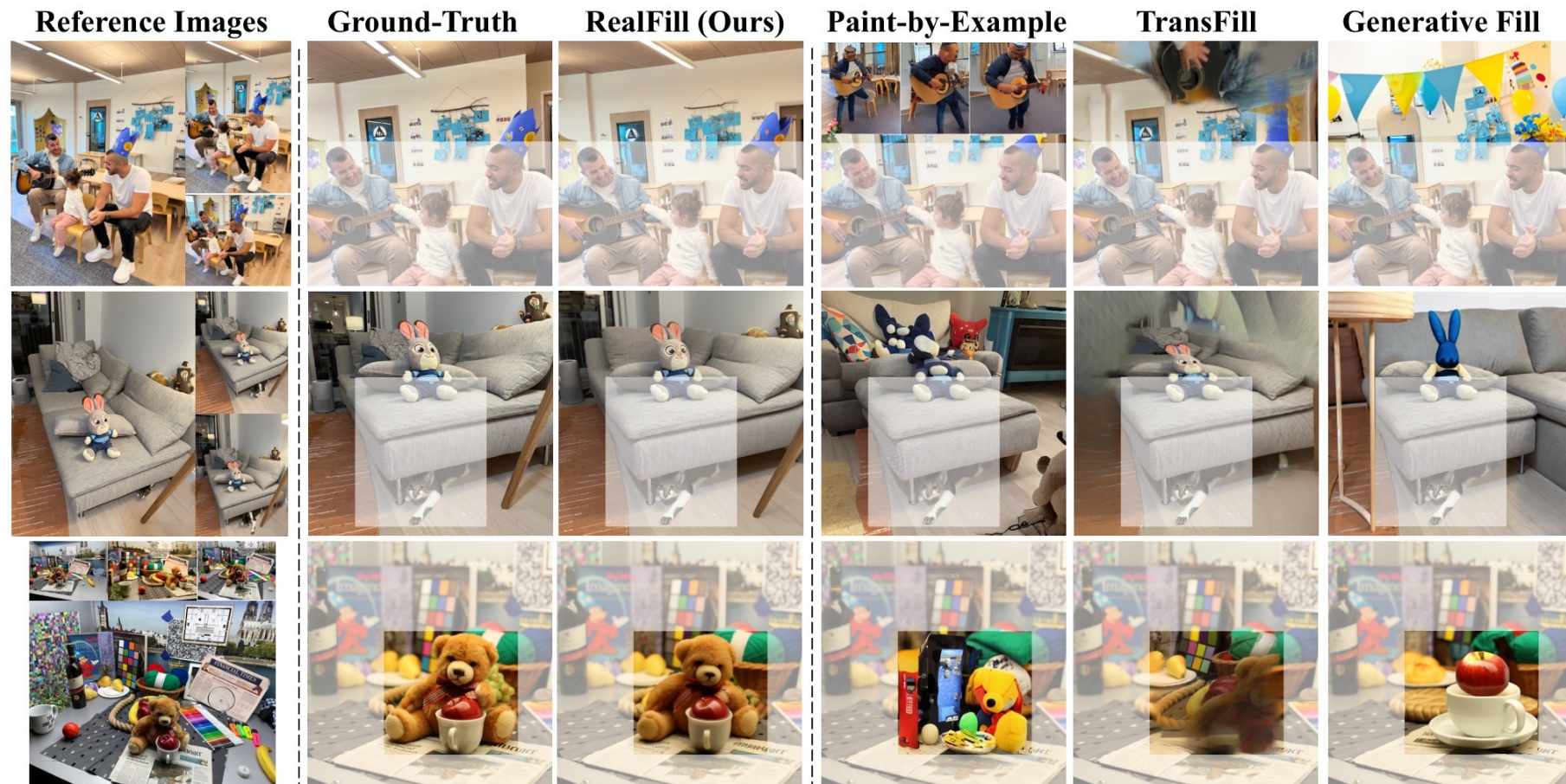
Tang et al., “RealFill: Reference-Driven Generation for Authentic Image Completion”, SIGGRAPH 2024.

Authentic Image Completion



RealFill 可以 inpaint 出與 references 一致的衣服、圖案與場景

Related Work



這個任務需要強大的 image prior + 專屬於這個場景的 knowledge

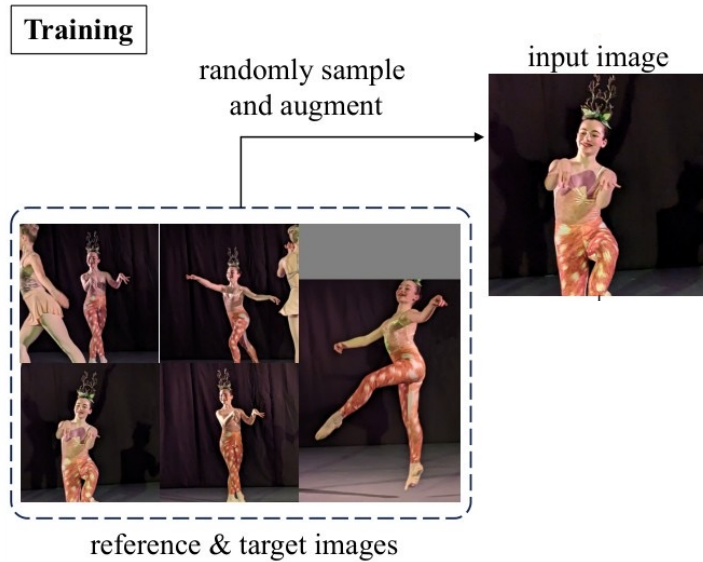
Method: Per-scene Training DreamBooth + LoRA on Inpainting Model

Training

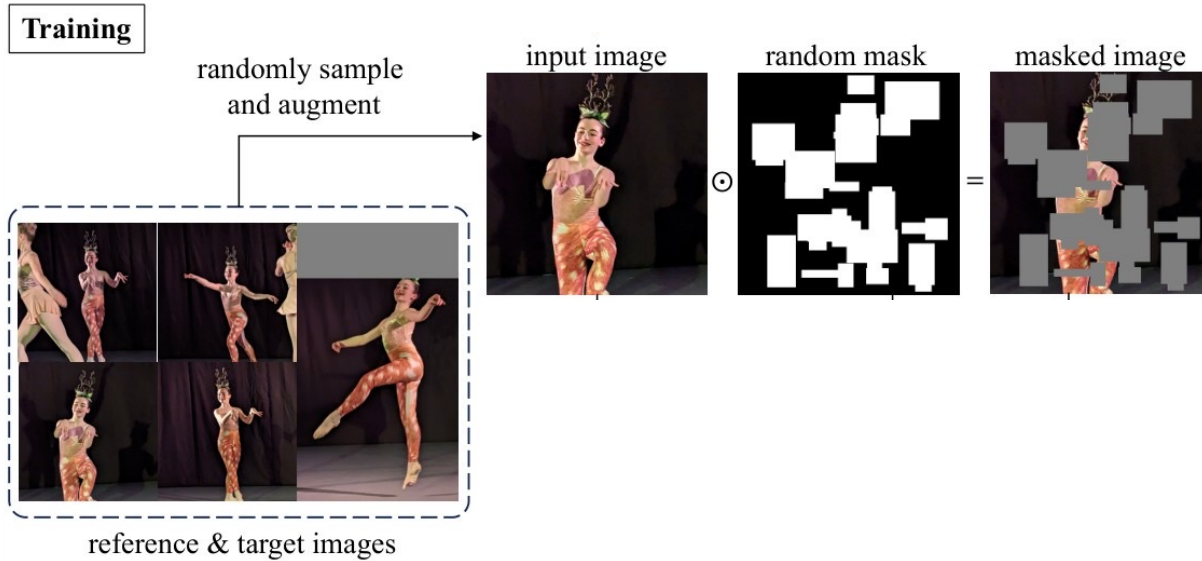


reference & target images

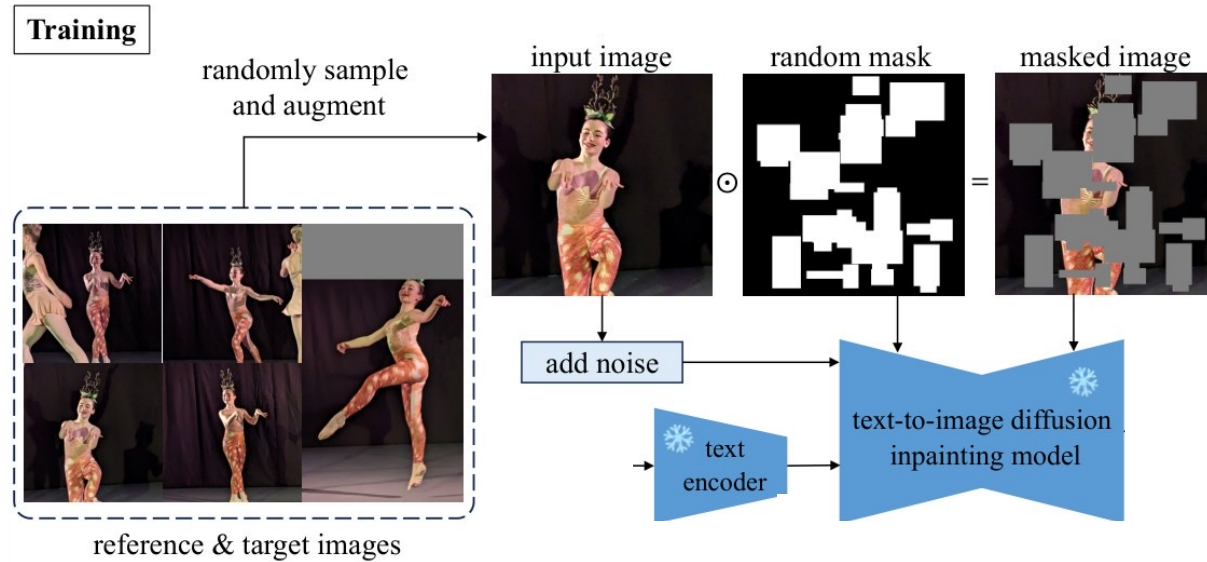
Method: Per-scene Training DreamBooth + LoRA on Inpainting Model



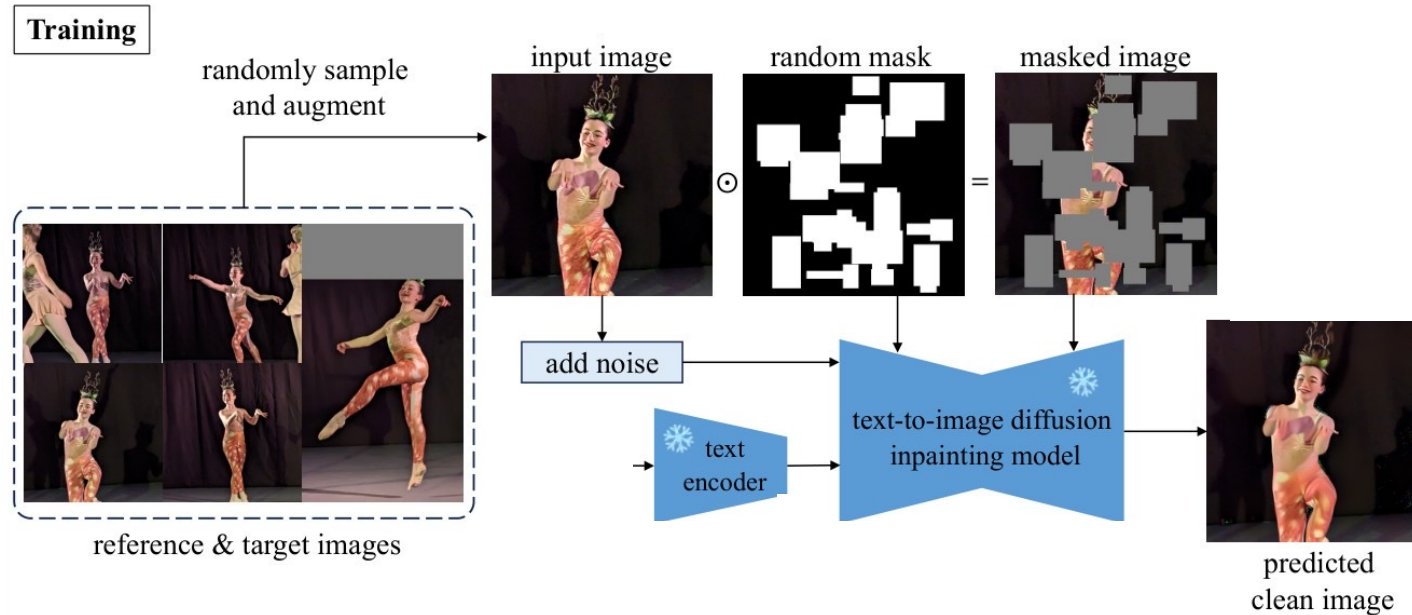
Method: Per-scene Training DreamBooth + LoRA on Inpainting Model



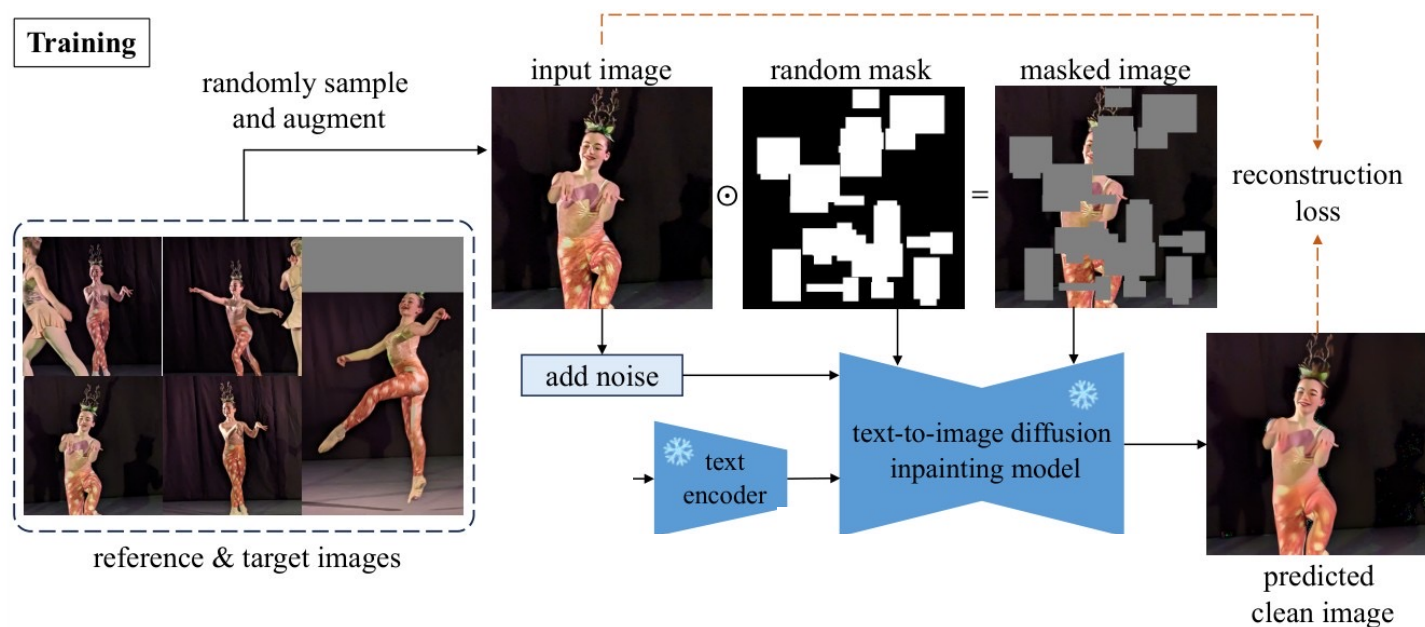
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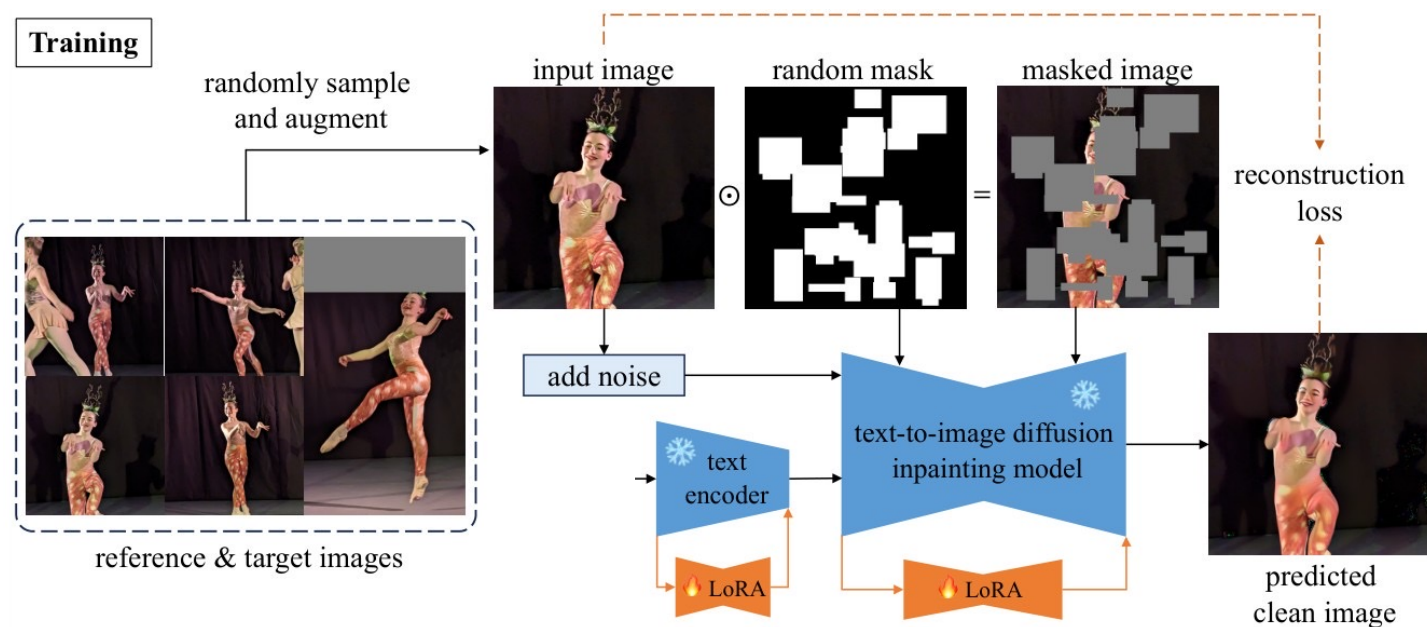


Method: Per-scene Training DreamBooth + LoRA on Inpainting Model



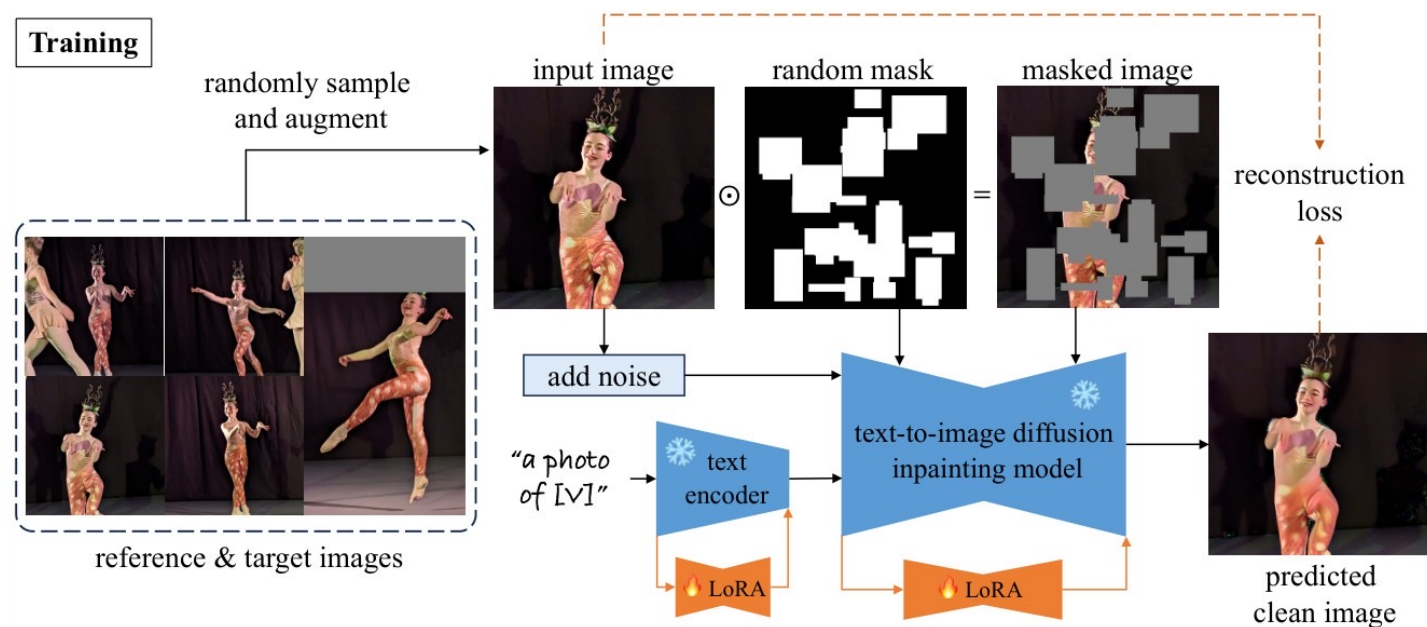
在 references 與 target 上隨機遮住區域讓模型 inpaint，並在已知區域計算 loss

Method: Per-scene Training DreamBooth + LoRA on Inpainting Model



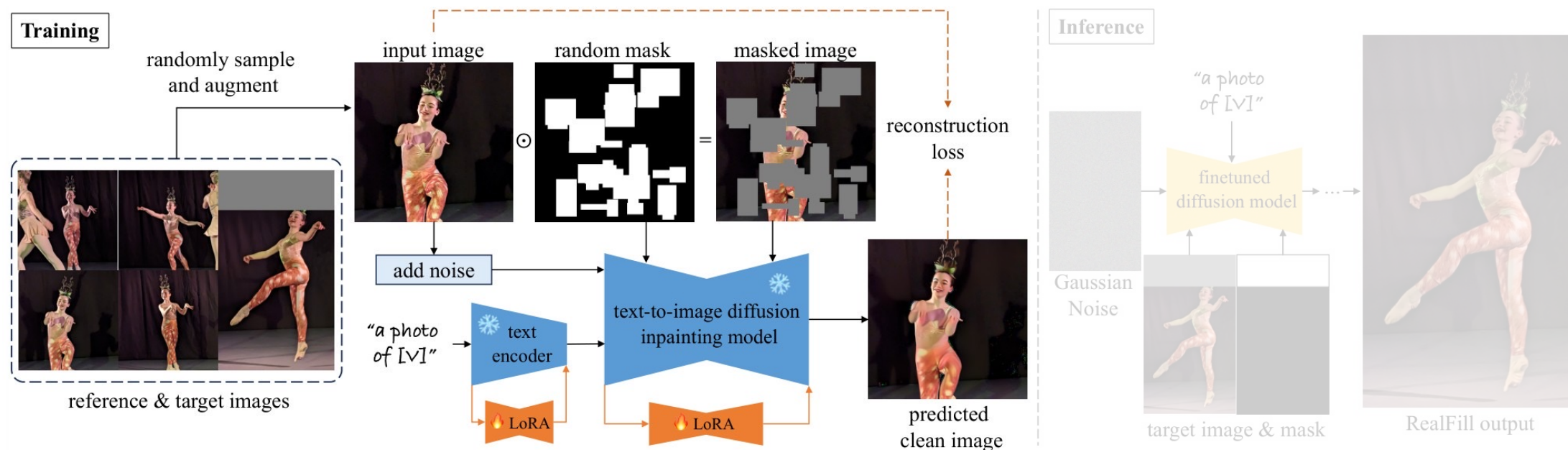
Freeze 住 Stable Diffusion 2 Inpainting
只在 text encoder 與 U-Net 加 LoRA

Method: Per-scene Training DreamBooth + LoRA on Inpainting Model



“a photo of [V]”，[V] 是稀有 token (與 DreamBooth 相同)

Method: Per-scene Training DreamBooth + LoRA on Inpainting Model

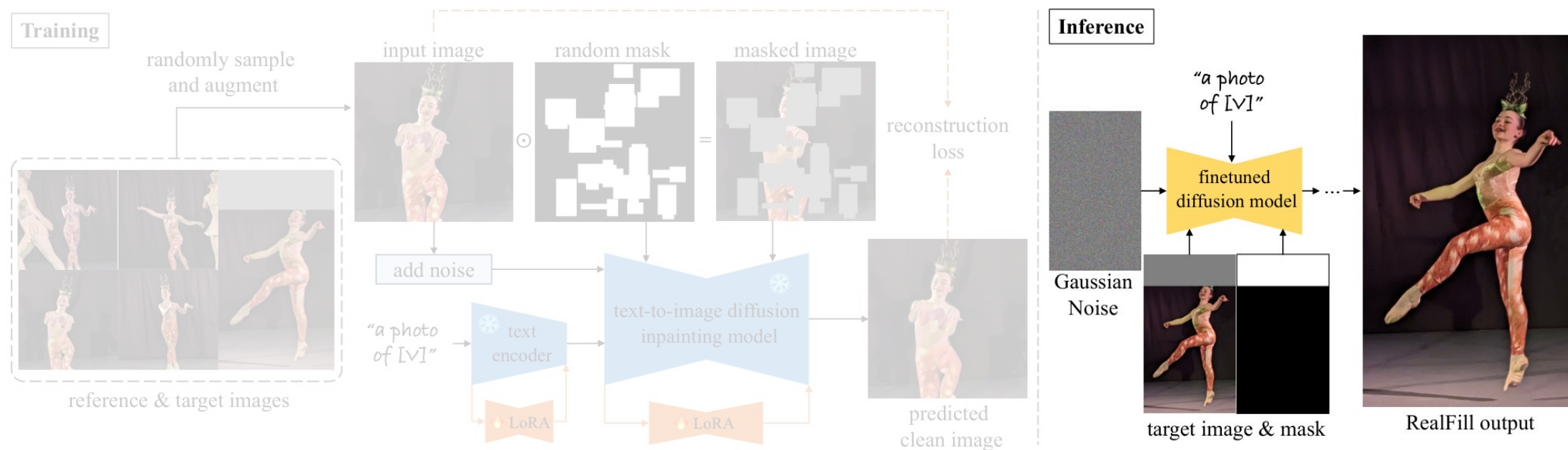


在 references + target 上 fine-tune
SD 2 Inpainting 的 LoRA

用隨機 mask inpainting 來訓練

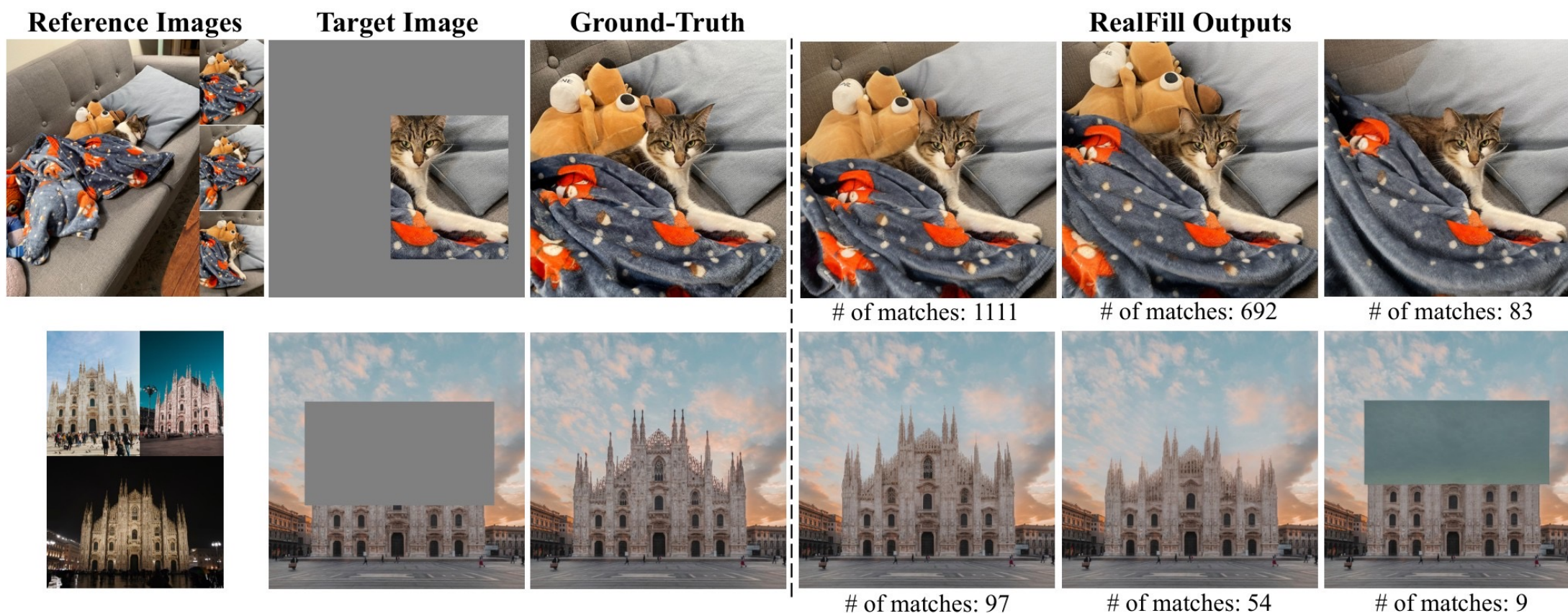
每個場景各自訓練，單張 A100 約 1 小時

Method: Inference



固定 prompt “a photo of [V]”，輸入 target 與 mask，用 fine-tune 後的模型 inpaint

Inference: Correspondence-Based Seed Selection



一次生成 64 張 -> 用 LoFTR (以 Transformer 做 local feature matching) 計算補洞區域與 references 的對應點數 -> 保留前 16 張，再由使用者挑選

Experiments: Outpainting

Reference Images



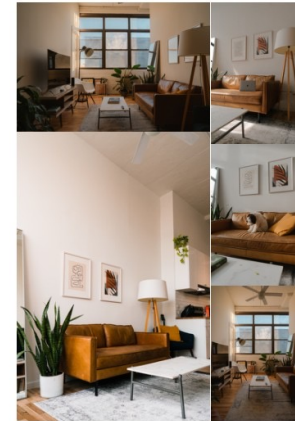
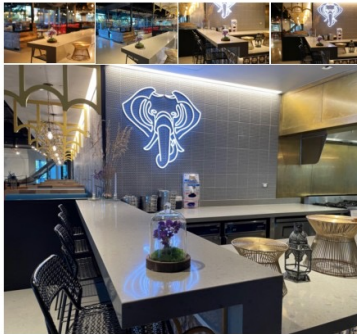
RealFill Outputs



Reference Images

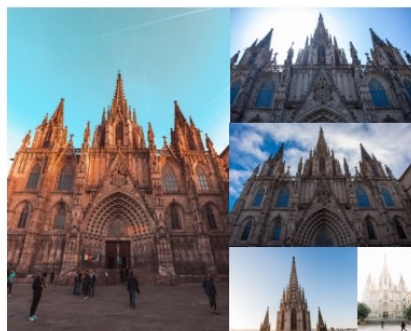


RealFill Outputs

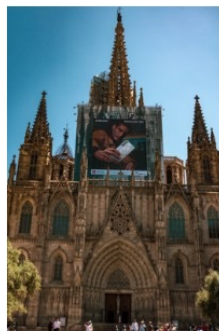


Experiments: Inpainting

Reference Images



Target Image



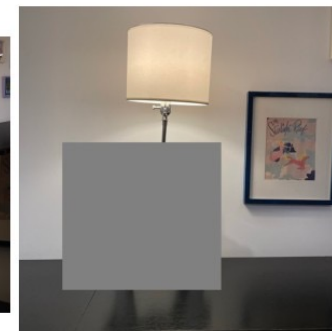
RealFill Output



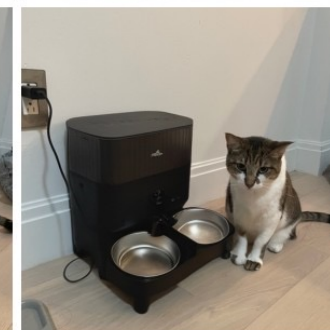
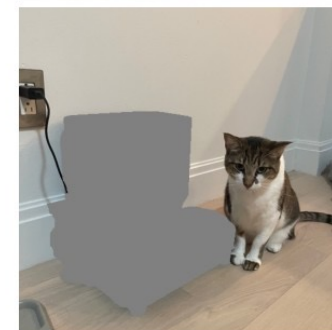
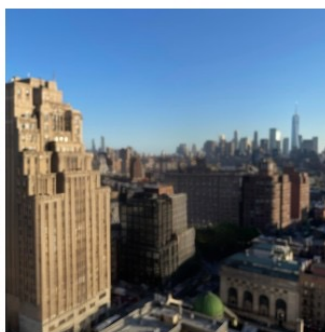
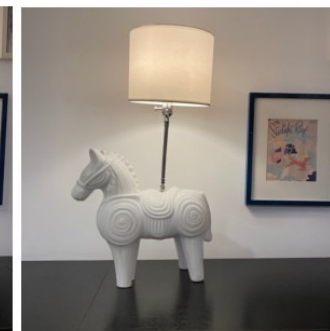
Reference Images



Target Image



RealFill Output



Experiments

Table 2. **Effect of correspondence-based seed selection.** It helps RealFill output higher quality results, i.e., filtering out samples with fewer matches results in better quantitative scores.

Filtering Rate	PSNR↑	SSIM↑	LPIPS↓	DreamSim↓	DINO↑	CLIP↑
0%	14.78	0.424	0.431	0.077	0.948	0.962
25%	15.01	0.427	0.421	0.066	0.955	0.967
50%	15.05	0.427	0.418	0.063	0.958	0.969
75%	15.10	0.427	0.417	0.060	0.961	0.970

Experiments

Table 3. **User study results.** Here we show the preference percentages for the most realistic and most faithful image completions across 58 scenes by 44 participants. RealFill significantly surpasses baselines, especially on the faithfulness criterion.

Method	Most Realistic $\uparrow$	Most Faithful $\uparrow$
TransFill [Zhou et al. 2021]	2.0%	3.7%
Paint-by-Example [Yang et al. 2023]	11.0%	1.9%
Generative Fill [Adobe Inc. 2023]	23.4%	7.2%
RealFill (ours)	63.7%	87.2%

Summary

- [2021.05] Classifier Guidance: 用 classifier 的 gradient 來做到 conditional generation
 - 要另外 train classifier，而且只能有固定的 class set
- [2021.12] Classifier-Free Guidance (CFG): 不需要 classifier
 - 只做到在 class label 上
- [2021.12] **GLIDE** (ICML 2022): CFG + Free-form Text Prompt，fine-tune text-guided inpainting
 - 在 pixel space 生成很慢，且只能用文字描述想要 inpaint 的區域（不能用 reference images）
- [2022] Latent Diffusion -> Stable Diffusion: Latent space 且 open source/weights，也有 finetune 一個 inpainting model: SD Inpainting
 - 一樣只能用文字描述想要 inpaint 的區域（不能用 reference images）
- [2022-23] DreamBooth + LoRA: Personalization 用幾張照片來記住特定主體
 - Generation not inpainting
- [2023.09] **RealFill** (SIGGRAPH 2024): 以 references inpaint 出真實的內容
 - 每個場景都要 finetune（約一小時），且大視角變化會失敗

Summary

- [2021.05] Classifier Guidance: 用 classifier 的 gradient 來做到 conditional generation
 - 要另外 train classifier，而且只能有固定的 class set
- [2021.12] Classifier-Free Guidance (CFG): 不需要 classifier
 - 只用在 text-to-image
- [2022.01] Inpainting Guidance: 用 inpainting 的 gradient 來做到 conditional generation
 - 只用在 inpainting
- [2022.02] Inpainting Guidance: 用 inpainting 的 gradient 來做到 conditional generation
 - 只用在 inpainting
- [2022.03] Inpainting Guidance: 用 inpainting 的 gradient 來做到 conditional generation
 - 只用在 inpainting
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 - 只用在 inpainting
- [2022.06] Inpainting Guidance: 用 inpainting 的 gradient 來做到 conditional generation
 - 只用在 inpainting
- [2022.07] Inpainting Guidance: 用 inpainting 的 gradient 來做到 conditional generation
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What's next?

Finetuning-free 的 personalization for generation/inpainting ?

結合 3D 或 video 的 prior ?